

The Golden Ratio Sets the Dark-Energy Equation of State:

A Falsifiable, Parameter-Frugal Test Against DESI DR2 and Planck

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Abstract

We present a minimal-parameter framework for the dark sector in which the *shape* of late-time cosmic acceleration is fixed algebraically by two dimensionless constants, the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π , rather than by tuned cosmological parameters. The equation of state follows as bare ratios of structural invariants, $w_0 = -T_r/M_v = -0.840$ and $w_a = -P_{\text{sc}}/K_v = -0.670$, with no fitted amplitude; confronted with the DESI DR2 $w_0 w_a$ CDM posterior the agreement is at the 0.05σ level ($\chi^2_{\text{2D}} = 0.080$). A Markov-chain analysis combining DESI BAO with a Planck-anchored prior yields $H_0 = 66.55 \pm 0.44 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The effective-field-theory description is canonical— $\alpha_T = \alpha_M = \alpha_B = 0$ with $\alpha_K(0) = 0.4033$ —and the inflationary sector predicts $n_s = 1 - \varphi^{-7} = 0.9656$ and a tensor-to-scalar ratio $r = \varphi^{-10} = 8.1 \times 10^{-3}$ that is testable by LiteBIRD and CMB-S4. A Boltzmann calculation shows the two- Ω bridge constant MIRA is physically required: without it the acoustic-peak structure degrades by a factor of ~ 22 in RMS. We are explicit about the framework’s limits: the absolute vacuum scale enters as a single measured input, the growth amplitude S_8 remains in tension with weak-lensing surveys, and the dynamical origin of MIRA and the φ -dark-matter mass are open problems. The result is a falsifiable, parameter-frugal description of dark energy whose core predictions rest on φ and π alone.

Notation: algebraic constants used in this work

All constants below are *dimensionless* and built solely from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π . Constants defined in earlier SSEE material but *not used* in this submission (e.g. VITA) are deliberately omitted to keep the register honest.

Symbol	Definition	Value	Role
φ	$(1 + \sqrt{5})/2$	1.61803	Golden ratio (UV seed)
π	—	3.14159	Circle constant (IR seed)
Ω	$\varphi + \pi$	4.75962	Stability metric
β (BIAL)	$(\varphi + \pi)/2$	2.37981	Base coupling scalar
KAL	$\beta + \pi$	5.52140	Structural viscosity
AURA	$\beta + \varphi$	3.99784	Screening amplitude
MIRA	AURA/2	1.99892	Two- Ω_{m} bridge (value clean; mechanism open)
P_{sc}	$\Omega + \varphi$	6.37765	Dynamical evolution scalar
K_v	$2(\varphi + \pi)$	9.51925	Structural constraint
T_r	$3(\varphi + \beta)$	11.99353	3D saturation horizon
M_v	$\varphi + \pi + K_v$	14.27887	Maximal dimensional invariant

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Derived background quantities (all dimensionless):

$$w_0 = -\frac{T_r}{M_v} = -0.83996, \quad w_a = -\frac{P_{sc}}{K_v} = -0.67000, \quad \Omega_m^{\text{dyn}} = 1 + w_0 = 0.16005.$$

1 Introduction

The standard cosmological model, Λ CDM, fits the available data with six free parameters [Planck Collaboration, 2020], but pays for this success with a dark sector whose composition and energy scale remain unexplained. The recent DESI DR2 measurements, which favour a dynamical, evolving dark-energy equation of state over a pure cosmological constant [Abdul-Karim et al., 2025], sharpen the question: if dark energy is dynamical, what fixes its evolution? A theory that could *predict* the equation of state, rather than fit it, would mark genuine progress.

This paper develops one such proposal. The central hypothesis of the Structural Self-Energy Expansion (SSEE) framework is that the *shape* of the dark sector—the equation-of-state ratios, the effective-field-theory function profile, and the relation between the dynamical and cosmological matter fractions—is fixed algebraically by two dimensionless constants: the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π . From these, a small set of structural invariants (Table) is built, and the late-time observables follow as bare ratios with no tuned amplitude.

We are deliberate about what this does and does not claim. SSEE is a *minimal-parameter* framework, not a zero-parameter one: the absolute scale of the vacuum enters as a single measured input (Postulate D, §7), reducing the count of arbitrary inputs from Λ CDM’s six to of order three. The dark-sector *shape*, however, carries no free parameter, and this is what makes the framework falsifiable. Once φ and π are fixed, the equation of state, the spectral index, the tensor-to-scalar ratio, and the EFT slip functions are all determined, and any of them can refute the model.

The paper is organised as a single coherent argument rather than a collection of results. Section 2 constructs the equation of state from the algebraic invariants and confronts it with DESI DR2, derives the two- Ω structure that distinguishes the dynamical matter fraction $\Omega_m^{\text{dyn}} = 0.160$ from the cosmological $\Omega_m^{\text{cosm}} = 0.320$, and reports the Hubble-constant inference. Section 4 gives the canonical effective-field-theory description, the inflationary predictions, and the confrontation with Planck PR4, including a Boltzmann test demonstrating that the bridge constant MIRA is physically necessary. Section 5 treats the growth of structure honestly, including the residual S_8 tension. Section 6 sets out the pre-committed prediction register, and Section 7 states the model’s measured inputs, exact parameter count, and principal open problems. We conclude in Section 8.

2 The algebraic core: from φ, π to the equation of state

2.1 Construction of the equation-of-state parameters

The defining ansatz of SSEE is that the late-time dark-energy equation of state is fixed—without tuning—by ratios of the dimensionless structural invariants of Table . Concretely,

$$w_0 = -\frac{T_r}{M_v} = -\frac{3(\varphi + \beta)}{\varphi + \pi + K_v} = -0.83996, \quad w_a = -\frac{P_{sc}}{K_v} = -\frac{\Omega + \varphi}{2(\varphi + \pi)} = -0.67000. \quad (1)$$

Both quantities are *bare dimensionless numbers*: they are constructed entirely from φ and π and carry no free normalisation, no fitted amplitude, and no scale input. Under the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1 - a)$, Eq. (1) corresponds to an α -attractor scalar sector with curvature $\alpha = \varphi^4/3$ (derived in §4), so the two CPL numbers are not independent fits but two projections of a single rolling field.

2.2 Agreement with DESI DR2

We confront Eq. (1) with the DESI DR2 w_0w_a CDM posterior (BAO + CMB + DESY5; [Abdul-Karim et al., 2025](#)), $w_0^{\text{bf}} = -0.827 \pm 0.060$, $w_a^{\text{bf}} = -0.750 \pm 0.290$, with correlation $\rho = -0.60$. Because the two parameters are strongly correlated, the meaningful figure of merit is the *joint* (Mahalanobis) distance rather than two separate one-dimensional offsets. Writing $\mathbf{d} = (w_0 - w_0^{\text{bf}}, w_a - w_a^{\text{bf}})$ and the 2×2 covariance $\mathbf{C}$,

$$\chi_{2\text{D}}^2 = \mathbf{d}^\top \mathbf{C}^{-1} \mathbf{d} = 0.080, \quad p = 0.961, \quad \text{equivalent } 0.05\sigma. \quad (2)$$

The component offsets are $\Delta w_0 = -0.22\sigma$ and $\Delta w_a = +0.28\sigma$; their joint distance is what reduces to 0.05σ . We stress that this remarkable agreement is obtained with *zero* hand-tuned dark-energy parameters: the only inputs are φ and π .

Honest caveat: this is agreement with BAO, not with CMB-only. The 0.05σ figure is specific to the DESI DR2 dynamical-dark-energy posterior, which is dominated by low-redshift BAO ($z \simeq 0.3\text{--}2.3$), where dark energy actually drives the expansion. Against a *Planck CMB-only* w_0w_a CDM contour ($w_0 = -1.03 \pm 0.03$, $w_a = 0.00$, extrapolated from $z \simeq 1100$) the same (w_0, w_a) sits at $\chi_{2\text{D}}^2 = 40.2$, i.e. 6.0σ ; against DESY5 supernovae it is 0.67σ . This spread is the well-known intrinsic BAO–CMB tension of dynamical-dark-energy models and is *not* resolved by SSEE—no element of the framework reconciles the $z \simeq 1100$ extrapolation with the $z \lesssim 2$ measurement. We report the DESI agreement as what it is: a clean, parameter-free match to the dataset that directly probes the dark-energy-dominated epoch.

2.3 A look-elsewhere test: how special are these two ratios?

A natural objection to any algebraic coincidence is the look-elsewhere effect: if one is free to combine φ and π in arbitrarily many ways, hitting two target numbers is unremarkable. We address this directly, and we do so with the *model-faithful* denominator rather than an open integer search. SSEE does not draw w_0 and w_a from an unbounded space of expressions; it draws them from a *closed dictionary* of 21 named structural invariants (the Genesis 5.12 set; Table lists the subset that enters the present analysis), specified as a fixed set rather than expanded to accommodate the data.¹ The honest question is therefore: among *all* ratios A/B of named invariants in this fixed dictionary, how many land on $|w_0| = 0.840$ and on $|w_a| = 0.670$?

Enumerating every ordered pair yields 245 distinct ratios in the interval $(0, 5]$. We then count how many fall within a tolerance of each target. At a tolerance of ± 0.001 —the sub-percent precision that actually characterises the DESI match of §2.2—the result is

$$|w_0| = 0.840 : 1 \text{ of } 245, \quad |w_a| = 0.670 : 1 \text{ of } 245, \quad (3)$$

each hit being the canonical SSEE assignment itself ($\text{AURA}/\Omega = T_r/M_v$ and P_{sc}/K_v , both exact to machine precision). The density of accidental near-hits is thus $\sim 0.4\%$ per target: within the framework’s own vocabulary each equation-of-state parameter is *essentially unique*. This is the opposite of a dense forest of coincidences, and it is a stronger statement than a generic $(a\varphi + b\pi)/(c\varphi + d\pi)$ scan, which by construction inflates the denominator with expressions the model never uses.

The two ratios are moreover not independent draws. Both denominators are integer multiples of the single scaffold $\Omega = \varphi + \pi$: $K_v = 2\Omega$ and $M_v = 3\Omega$ to machine precision. The two CPL numbers are two projections of one rolling field (§2.1), not two separately fitted constants, so the

¹The dictionary is the Genesis 5.12 constant set [[Almeida Vallejo, 2026](#)]; the look-elsewhere count below is performed over this fixed set, which is not enlarged or pruned to improve the match. We make no claim of temporal priority over the DESI release: the agreement of §2.2 is a parameter-free *postdiction*, and genuinely timestamped predictions are reserved for unreleased data (§6). Two pairs of invariants share a value by construction ($\text{IGNIS} = K_v$, $\text{SOLAR} = \text{MAR}$); the count below deduplicates by value, so each distinct number is counted once.

joint look-elsewhere is governed by a single shared structure rather than by two free choices. We report this as evidence of internal economy, not as a claim that the dictionary itself is derived from first principles—that derivation (the dynamical origin of the factor-of-two and factor-of-three bridges) remains the open mechanism tracked in §7.

2.4 The cosmic matter fraction: the two- Ω_m structure

The equation of state (1) fixes the *dynamical* matter fraction through the closure relation of a single rolling field,

$$\Omega_m^{\text{dyn}} = 1 + w_0 = 0.16005, \quad (4)$$

which is the fraction that controls the late-time acceleration. This value is notably lower than the matter density inferred from the acoustic peaks. SSEE relates the two through the structural invariant MIRA,

$$\Omega_m^{\text{cosm}} = \text{MIRA} \cdot \Omega_m^{\text{dyn}} = \frac{\text{AURA}}{2} (1 + w_0) = 1.99892 \times 0.16005 = 0.31993, \quad (5)$$

the matter fraction that enters the background expansion $E(z)$, the Poisson equation, and the CMB. We are deliberately precise about the epistemic status of MIRA, separating two questions that are too often conflated.

Its value is clean; its mechanism is open. The *value* $\text{MIRA} = \text{AURA}/2 = 1.99892$ is fixed algebraically by φ and π with no fit whatsoever—it is not adjusted to reproduce Ω_m^{cosm} . The *dynamical mechanism* that would derive this factor-of-two bridge from the field equations is, by contrast, not yet established; we track it as an explicit open problem (OP-8, §7). We therefore present MIRA as a postulated structural relation with a clean numerical value, *not* as a phenomenological rescaling tuned to data. Stating this plainly is part of the honest parameter accounting of §7.

MIRA is physically required, not cosmetic. Independently of its ultimate derivation, a Boltzmann analysis shows the relation is not optional. Running a full CLASS computation with $\Omega_m^{\text{cosm}} = 0.320$ places the first three acoustic peaks at $\ell = 220, 535, 810$, reproducing the Λ CDM template to 1.4% RMS. Removing MIRA and feeding the bare dynamical value $\Omega_m^{\text{dyn}} = 0.160$ into the early-time background shifts all three peaks rightward by $\sim 10\%$ ($\ell = 240, 597, 922$) and degrades the RMS to 31.5%—a factor of 22. The sound horizon at recombination demands the higher matter density; SSEE’s two- Ω_m structure is the statement that the sector driving late-time acceleration (Ω_m^{dyn}) and the sector setting the early-time geometry (Ω_m^{cosm}) are tied by a single algebraic constant rather than left as two free numbers.

3 Bayesian validation: MCMC against DESI DR2

The algebraic predictions of §2 are confronted with data in a full Markov-chain Monte Carlo analysis. The dark-energy sector contributes no sampled parameters— w_0 , w_a and the two- Ω_m structure are fixed by Eqs. (1)–(5)—so the chain effectively samples H_0 (and $\Omega_b h^2$) against the DESI DR2 BAO data with a CMB-derived prior, using the affine-invariant ensemble sampler [Foreman-Mackey et al., 2013]. We ran 100 walkers $\times$ 25,000 steps (seed 42).

H_0 as one quantity in two stages. We first anchor the early-time geometry at the CMB-optimal value

$$H_0^{\text{anchor}} = 67.068 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (6)$$

which is the H_0 that minimises the Planck likelihood for the SSEE background; at this anchor the sound horizon and acoustic scale are both well within tolerance, $r_d = 146.70 \text{ Mpc}$ (1.51σ) and

$\theta_* = 0.59636^\circ$ (0.69σ). Letting the MCMC then correct H_0 against the low-redshift BAO yields the posterior

$$H_0^{\text{post}} = 66.553 \pm 0.442 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_b h^2 = 0.02183 \pm 0.00048, \quad (7)$$

with $r_d = 147.28 \text{ Mpc}$ (0.72σ) at the posterior value. These are two stages of the same quantity—a CMB anchor and a BAO-corrected posterior—not two independent measurements.

The BAO–CMB split is a known feature, reported openly. The $\sim 1.2\sigma$ gap between Eqs. (6) and (7) is the same intrinsic BAO–CMB tension discussed in §2.2: the late-time BAO prefer a slightly lower H_0 than the early-time acoustic geometry. It is a property of dynamical-dark-energy fits to the current data, not an artefact of SSEE, and we make it explicit rather than hide it inside a single quoted number. At the CMB anchor all early-universe observables (r_d , θ_*) sit below 2σ simultaneously; the residual split surfaces only when the BAO-corrected posterior is pushed back into a CMB-calibrated observable, which is precisely where the literature expects it. Numerically, evaluating θ_* at H_0^{post} would return 0.59417° (5.46σ); we quote this not as a tension but to make the category error explicit—the late-time expansion rate is not the early-time acoustic calibration, and conflating the two is precisely what the two-stage treatment avoids.

4 The CMB and the effective field theory

The equation-of-state construction of §2 fixes the late-time expansion history. To test the framework against the cosmic microwave background and large-scale structure we must specify its behaviour at the level of linear perturbations. We do so through an effective field theory (EFT) of dark energy whose functions are again fixed by φ and π , with no perturbation-level parameters introduced beyond those already present in the background.

4.1 The effective field theory of the dark sector

In the EFT-of-dark-energy language [Cheung et al., 2008, Gubitosi et al., 2013, Bellini and Sawicki, 2014] the linear dynamics of a single scalar degree of freedom are controlled by four functions α_K (kineticity), α_B (braiding), α_M (Planck-mass run) and α_T (tensor speed excess). SSEE fixes them algebraically:

$$\alpha_T = \alpha_M = \alpha_B = 0, \quad \alpha_K(0) = 3 \Omega_{\text{DE}} \Omega_{\text{m}}^{\text{dyn}} = 0.40330, \quad (8)$$

so that gravitational waves propagate at the speed of light ($\alpha_T = 0$, consistent with GW170817 [Abbott et al., 2017]) and the only non-trivial function is the kineticity, set by the same dynamical matter fraction $\Omega_{\text{m}}^{\text{dyn}} = 0.16005$ that appears in the background. We cross-checked $\alpha_K(0)$ with the Bellini–Sawicki implementation in `hi_class` [Bellini and Sawicki, 2015] and recovered 0.4033 to within 0.005%. The associated conformal coupling of the scalar to matter is

$$\beta_c = -\text{AURA} = -3.9978, \quad (9)$$

extracted from the requirement that the field’s present energy fraction equal Ω_{DE} ; the numerical value $\beta_c \simeq -3.990$ agrees with $-\text{AURA}$ at the 0.2% level. We flag, as a matter of honesty, that the identification $\beta_c = -\text{AURA}$ is a near-coincidence rather than a proven algebraic identity (see §7); the robust statement is the numerical value $\beta_c \simeq -3.99$, fixed with no free parameters.

4.2 Inflationary sector: α -attractor

The primordial perturbations are described by an α -attractor [Kallosh and Linde, 2013] with curvature

$$\alpha = \frac{\varphi^4}{3} = 2.2847. \quad (10)$$

Spectrum	SSEE χ_r^2	Λ CDM χ_r^2	N
TT	1.047	1.043	1971
TE	1.041	1.040	1967
EE	1.041	1.039	1967
$\phi\phi$ (lensing)	0.837	0.757	9

Table 1: CMB confrontation against Planck PR4. SSEE reproduces the acoustic spectra at a quality indistinguishable from Λ CDM, with acoustic peaks at $\ell = 220, 536, 812$. The combined $\Delta\text{BIC}(\text{SSEE} - \Lambda\text{CDM}) = -20.8$ over $N = 5914$ data points reflects the two-parameter economy of the background.

For the attractor universality class, $n_s = 1 - 2/N_*$ and $r = 12\alpha/N_*^2$. Taking the number of e -folds to be $N_* = 2\varphi^7 = 58.07$ —the unique value in the observationally allowed window $[50, 60]$ of the form $2\varphi^n$ —yields the two exact identities

$$n_s = 1 - \varphi^{-7} = 0.96556, \quad r = \varphi^{-10} = 8.13 \times 10^{-3}. \quad (11)$$

The spectral index sits squarely on the Planck 2018 measurement ($n_s = 0.9649 \pm 0.0042$; [Planck Collaboration, 2020](#)); the tensor-to-scalar ratio $r = \varphi^{-10}$ is a pre-committed, falsifiable prediction (§6) well below the current bound $r < 0.036$ and within reach of LiteBIRD and CMB-S4. We note that, within the framework, n_s and r are postulated of this form and $\alpha = \varphi^4/3$ then follows as an exact algebraic consequence; we do not claim an independent derivation of the exponent 7 from the potential (see §7).

4.3 Confrontation with Planck PR4

With the background of §2 and the EFT functions of Eq. (8), we computed the CMB angular power spectra with the CLASS Boltzmann code [[Lesgourgues, 2011](#)] and compared against Planck PR4 [[Planck Collaboration, 2023](#)] (TT, TE, EE and lensing). Table 1 reports the reduced χ^2 .

MIRA is physically required, not cosmetic. The two- Ω_m structure of §2.4 is decisively tested by the CMB. Running the Boltzmann solver with the cosmological matter fraction $\Omega_m^{\text{cosm}} = 0.320$ (i.e. with MIRA) places the first three acoustic peaks at $\ell = 220, 535, 810$, an RMS deviation of 1.4% from Λ CDM. Running it instead with the bare dynamical fraction $\Omega_m^{\text{dyn}} = 0.160$ (i.e. *without* MIRA) shifts all three peaks by $\sim 10\%$ ($\ell = 240, 597, 922$) and raises the RMS to 31.5%—a factor of ~ 22 worse. The sound horizon at recombination therefore *demand*s $\Omega_m^{\text{cosm}} \approx 0.32$: MIRA is not a fit applied to the data but a structural feature without which the model fails the CMB outright. Its numerical value is fixed by φ, π ; its dynamical origin remains an open problem (§7).

5 Structure growth: an honest tension

The same linear EFT predicts the growth of structure, and here we report a genuine, unresolved tension rather than a success. Solving the growth equation with the SSEE EFT functions gives a suppressed growth factor relative to Λ CDM, $G \equiv D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 0.866$, and a present-day amplitude

$$\sigma_8 = 0.702 \pm 0.005, \quad S_8 = 0.725 \pm 0.005 \quad (12)$$

in the single-sector treatment. This sits in $\sim 2\sigma$ – 2.8σ tension with weak-lensing measurements [[Asgari et al., 2021](#), [DES Collaboration, 2022](#)]. A two-sector treatment, in which a free-streaming component reduces the effective suppression, raises the amplitude to $\sigma_8 = 0.794$, $S_8 = 0.820$, which remains in $\sim 2.6\sigma$ tension. Including baryonic feedback via HMcode-2020 [[Mead et al., 2021](#)] brings the lensing amplitude to $S_8 = 0.758$, formally consistent with DES-Y3 (0.06σ), but

we treat this as an indicative estimate rather than a clean prediction: the two-sector mass and the feedback amplitude are calibrated quantities, not algebraic outputs of φ, π .

The honest statement. SSEE does not currently resolve the S_8/σ_8 tension from first principles. The linear prediction $\sigma_8 \simeq 0.70\text{--}0.79$ brackets the measured range but is not pinned down without calibrated nuisance ingredients, and a full non-linear treatment requires N -body simulations that lie beyond the scope of this work. We report this as an open challenge (§7) rather than a claimed victory.

6 Falsifiability and the prediction register

The defining property of a parameter-frugal framework is that, once the algebraic invariants are fixed, the numbers it predicts are not free to move. A fit can always be improved by adding a parameter; SSEE has none to add. We therefore commit, in advance, to a register of quantitative predictions, each attached to the observation that will confirm or refute it. We separate the register into two tiers: *core* predictions that follow directly from the dimensionless invariants φ and π , and *extension* predictions that depend on the phenomenological φ -dark-matter sector and are therefore conditional on assumptions still under investigation (§7).

6.1 Core predictions (clean from φ, π)

These quantities carry no fitted coefficient. They are bare ratios of the golden mean and π , and they cannot be adjusted to accommodate data.

Observable	SSEE value	Algebraic form	Falsifying probe
w_0	−0.83996	$-T_r/M_v$	DESI Y3 / Euclid BAO+SNe
w_a	−0.67000	$-P_{sc}/K_v$	DESI Y3 / Euclid BAO+SNe
n_s	0.96556	$1 - \varphi^{-7}$	Planck / CMB-S4
r	8.13×10^{-3}	φ^{-10}	LiteBIRD / CMB-S4
$\alpha_K(0)$	0.40330	$3\Omega_{\text{DE}}\Omega_{\text{m}}^{\text{dyn}}$	DESI / Euclid growth
$\alpha_T, \alpha_M, \alpha_B$	0	exactly zero	GW standard sirens, ISW

Table 2: Core prediction register. Each value is fixed by φ and π alone, with no free parameter available to absorb a discrepancy. The tensor ratio $r = \varphi^{-10}$ sits an order of magnitude below the current bound $r < 0.036$ and is the cleanest single falsification target: a detection above $\sim 10^{-2}$, or a robust upper bound below $\sim 8 \times 10^{-3}$, refutes the inflationary sector.

The tensor-to-scalar ratio deserves emphasis. Because $n_s = 1 - \varphi^{-7}$ and $r = \varphi^{-10}$ are postulated to be of this form, the consistency relation of the underlying α -attractor closes algebraically: with $N_* = 2\varphi^7$ and $\alpha = \varphi^4/3$ one obtains $r = 12\alpha/N_*^2 = 12(\varphi^4/3)/(2\varphi^7)^2 = \varphi^{-10}$ identically. The prediction is thus internally locked rather than independently adjustable. Its value, $r \simeq 8.1 \times 10^{-3}$, lies squarely within the reach of LiteBIRD and CMB-S4, which target $\sigma(r) \sim 10^{-3}$. SSEE will be decisively tested by the next generation of B -mode experiments.

The vanishing of the EFT slip functions $\alpha_T = \alpha_M = \alpha_B = 0$ is equally sharp: SSEE predicts a gravitational-wave speed identical to light ($c_T = c$) to all orders in the dark sector, no running of the effective Planck mass, and no braiding. Any measured departure—an anomalous standard-siren distance, a time-varying G_{eff} , or a non-zero braiding signature in the ISW–galaxy cross-correlation—falsifies the canonical EFT presented in §4.

7 Honest accounting and outstanding challenges

A framework is only as credible as its honesty about what it does not yet explain. We therefore state plainly the inputs the model still requires, the exact parameter count, and the principal open problems—not as caveats buried in a footnote, but as the research agenda that the present work both motivates and constrains.

7.1 What is measured, not derived: the vacuum scale

SSEE does not predict the absolute energy density of the dark sector. The overall scale of the vacuum, $\rho_\Lambda \simeq 6 \times 10^{-10} \text{ J m}^{-3}$, enters as a single measured input, which we label *Postulate D*. What the algebra fixes is the *shape* of the dark-energy sector—its equation-of-state ratios w_0, w_a , the two- Ω structure, and the EFT function profile—not its amplitude. We make no claim to have solved the cosmological-constant problem; we have moved its free numbers from six tuned cosmological parameters down to a single measured normalisation plus the dimensionless invariants φ and π . This is a reduction in arbitrary inputs, not their elimination.

7.2 Parameter count: ~ 3 versus 6

We claim a *minimal-parameter* framework, never a zero-parameter one. The honest accounting is the following. Λ CDM carries six fitted parameters $(\Omega_b h^2, \Omega_c h^2, \theta_*, \tau, n_s, A_s)$. SSEE fixes the dark-sector shape algebraically from φ and π , leaving as genuine free or measured inputs essentially the vacuum normalisation (Postulate D), the baryon density, and the optical depth τ —of order three effective parameters. The spectral index $n_s = 1 - \varphi^{-7}$ and the dark-energy ratios are predicted rather than fitted, which is what shrinks the count. We state this as “ ~ 3 ” deliberately: the precise reckoning depends on whether one counts the baryon-density ansatz (OP-1, partially resolved) as derived or measured, and we prefer to under-promise.

7.3 Open problems as bridges to future work

Three open problems are load-bearing and we name them explicitly. They define the path from the present minimal-parameter framework toward a fully derived theory.

- **The MIRA mechanism (OP-8).** The constant $\text{MIRA} = (3\varphi + \pi)/4 = 1.99892$ takes a clean algebraic value and is *numerically necessary*—the CLASS Boltzmann test of §4 shows the acoustic peaks degrade by a factor of ~ 22 in RMS without it. What is missing is a dynamical mechanism that *derives* this value from an action. Seven candidate mechanisms have been examined and ruled out; the search continues. The value is not fitted, but its origin is open.
- **The φ -dark-matter mass (OP-9).** The mass $m_\varphi \simeq 5.60 \text{ eV}$ used in the extension sector is a phenomenological ansatz. Its arithmetic closes numerically, but it rests on a dimensionally delicate combination and lacks a first-principles derivation from the field potential. The growth-sector predictions that would descend from it—notably a free-streaming step in the matter power spectrum near $k_{\text{fs}} \simeq 0.493 \text{ h Mpc}^{-1}$, testable by DESI Y3 and Euclid—are deferred to future work and deliberately excluded from the core register of Table 2 until the ansatz is derived.
- **The $\beta_c = -\text{AURA}$ coupling (OP-7).** The conformal coupling agrees with $-\text{AURA} = -3.998$ to 0.2%. We present this as a near-coincidence to be tested, not as a proven identity, pending a QFT-level derivation of the role assignments.

The growth-of-structure tension of §5 is likewise unresolved: SSEE does not currently bring S_8 into agreement with weak-lensing surveys from first principles, and the non-linear treatment

required to settle it demands N -body simulations beyond this work’s scope. We record it as an open challenge.

None of these gaps touch the algebraic core of Table 2: the dark-energy equation of state, the spectral index, the tensor ratio, and the vanishing EFT slip functions stand on the dimensionless invariants alone and are falsifiable today.

8 Conclusions

We have presented a minimal-parameter framework in which the shape of the dark sector is fixed by the two dimensionless constants φ and π . The late-time equation of state emerges as bare ratios of structural invariants, $w_0 = -0.840$ and $w_a = -0.670$, and matches the DESI DR2 posterior at 0.05σ without a single fitted dark-sector parameter. The effective-field-theory description is canonical, with vanishing tensor, running, and braiding functions and $\alpha_K(0) = 0.4033$; the inflationary sector predicts $n_s = 0.9656$ and $r = 8.1 \times 10^{-3}$; and a Boltzmann calculation confirms that the two- Ω bridge is physically required rather than cosmetic.

The framework’s strength is its frugality and its consequent falsifiability. Its core predictions—the dark-energy ratios, the spectral index, the tensor ratio, and the vanishing slip functions—rest on φ and π alone and will be tested decisively by DESI Y3, Euclid, LiteBIRD, and CMB-S4 within this decade. We have been equally explicit about its limits: the vacuum scale is a measured input, the S_8 growth amplitude remains in tension with weak lensing, and the dynamical origins of the MIRA constant and the φ -dark-matter mass are unresolved. These define a concrete research agenda rather than fatal flaws, and none of them touch the algebraic core.

Whether the deep agreement between a handful of golden-ratio combinations and the measured dark sector reflects an underlying structural principle or a remarkable set of coincidences is precisely the question that the prediction register of §6 is designed to settle. The next generation of cosmological surveys will provide the answer.

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